DESCENT OF THE FLAT-AT-PRIMES-OVER-AN-IDEAL CONDITION ALONG A DIRECTED COLIMIT OF RINGS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Finite presentation is preserved by base change, so the base changed module is still finitely presented. Algebra, Lemma \ref{algebra-lemma-base-change-finiteness}, cited at proof lines 4161-4163. Makes openness of the flat locus available for the base changed module.
- The flat locus of a finitely presented module is open. Algebra, Theorem \ref{algebra-theorem-openness-flatness}, cited at proof lines 4164-4165. Produces finitely many basic open sets covering the closed set cut out by the ideal.
- Finiteness of a flat module is preserved under passing to a sufficiently large index in the colimit. Algebra, Lemma \ref{algebra-lemma-flat-finite-presentation-limit-flat}, cited at proof lines 4188-4189. Shows that the localised modules become flat for large enough indices in the system.

Source reference. Original lemma 05LM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7241–7241. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 052W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6584–6584. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05G5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7049–7049. Navigation pointer to a statement the

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Source reference. Original lemma 00RC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 998–998. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02JO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3343–3343. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $R \to S$ be a ring map of finite presentation. Let M be an S-module of finite presentation. Let $R' = \operatorname{colim}_{\lambda\in\Lambda} R_\lambda$ be a directed colimit of R-algebras. Let $I_\lambda \subset R_\lambda$ be ideals such that $I_\lambda R_\mu \subset I_\mu$ for all $\mu \geq \lambda$ and set $I' = \operatorname{colim}_\lambda I_\lambda$. If (052W) holds for $(R' \to S \otimes_R R', I', M \otimes_R R')$, then there exists a $\lambda \in \Lambda$ such that (052W) holds for $(R_\lambda \to S \otimes_R R_\lambda, I_\lambda, M \otimes_R R_\lambda)$.*

Proof. We are going to write $S_\lambda = S \otimes_R R_\lambda$, $S' = S \otimes_R R'$, $M_\lambda = M \otimes_R R_\lambda$, and $M' = M \otimes_R R'$. The base change S' is of finite presentation over R' and M' is of finite presentation over S' and similarly for the versions with subscript λ, see Algebra, Lemma 05G5. By Algebra, Theorem 00RC the set

$$U' = \{\mathfrak{q}' \in \operatorname{Spec}(S') \mid M'_{\mathfrak{q}'} \text{ is flat over } R'\}$$

is open in $\operatorname{Spec}(S')$. Note that $V(I'S')$ is a quasi-compact space which is contained in U' by assumption. Hence there exist finitely many $g'_j \in S'$, $j = 1, \ldots, m$ such that $D(g'_j) \subset U'$ and such that $V(I'S') \subset \bigcup D(g'_j)$. Note that in particular $(M')_{g'_j}$ is a flat module over R'.

We are going to pick increasingly large elements $\lambda \in \Lambda$. First we pick it large enough so that we can find $g_{j,\lambda} \in S_\lambda$ mapping to g'_j. The inclusion $V(I'S') \subset \bigcup D(g'_j)$ means that $I'S' + (g'_1, \ldots, g'_m) = S'$ which can be expressed as $1 = \sum z_s h_s + \sum f_j g'_j$ for some $z_s \in I'$, $h_s, f_j \in S'$. After increasing λ we may assume such an equation holds in S_λ. Hence we may assume that $V(I_\lambda S_\lambda) \subset \bigcup D(g_{j,\lambda})$. By Algebra, Lemma 02JO we see that for some sufficiently large λ the modules $(M_\lambda)_{g_{j,\lambda}}$ are flat over R_λ. In particular the module M_λ is flat over R_λ at all the primes lying over the ideal I_λ. □